Unit 5 – Polynomial Functions – Day 9 *Accelerated II/III*

Analyzing Graphs of Polynomial Functions

End Behavior

$x \to -\infty$ *as* $f(x) \to$ _____ (left side of the graph)

$x \to +\infty$ *as* $f(x) \to$ _____ (right side of the graph)





$f(x) = a^n x^n + \ldots$	$a_n > 0$	$a_n < 0$
n is even		
n is odd		

Example 1: Determine the end behavior of the following polynomials:

A) $f(x) = x^3 + 2x^2 - 3x - 2$

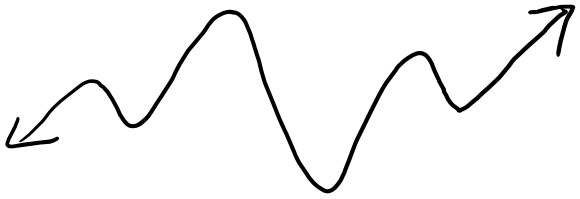
B) $f(x) = -24x^{10} - \frac{3}{16}x^4 + 4x$

Turning Points – The graph of every polynomial function of degree n has at most _____ turning points. Moreover, if a polynomial function has _____distinct real zeros, then its graph has exactly _____ turning points.

Maximum/Minimum Points

Relative/Local Maximum: The y-coordinate of a turning point is higher than all nearby points.

Relative/Local Minimum: The y-coordinate of a turning point is lower than all nearby points.



Zero, solutions, factors, x-intercepts: Let $f(x) = a_n x^n + \ldots. + a_1 x + a_0$ be a polynomial function and *k* is a real number, the following statements are equivalent

Zero: _____is a zero of the polynomial f

Solution: _____is a solution of the polynomial equation $f(x) = 0$

Factor: _____is a factor of the polynomial $f(x)$

X-intercept: $(___, 0)$ is an x-intercept of the graph of the polynomial f

Example 2: Find the x-intercepts of the graph of $f(x) = x^3 - x^2 - x + 1$

$f(x) = x^3 - x^2 - x + 1$ x-intercepts are _____

**Note: In the above example $x = 1$ is a "repeated zero"

Multiplicity Rules: For an equation $f(x)=0$, k is a repeated solution if and only if the factor $(x-k)$ has a degree greater than 1 when $f(x)$ is factored completely (AKA: the number of times a zero is repeated)

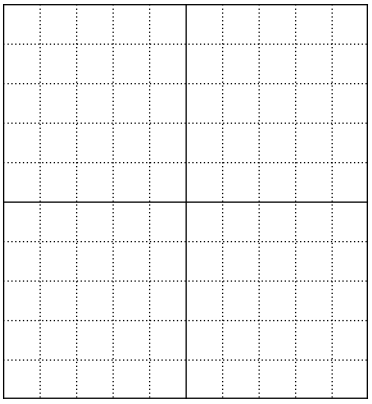
$(x-k)^n$, $n>1$ yields a repeated zero of $x=k$ with multiplicity of n.

- If n is odd, then graph will cross through the x-axis at k.
- If n is even, then the graph will only touch the x-axis at k and "bounce" back.

i.e. $f(x)=(x-2)(x-4)^2(x+1)^5$

We can graph polynomial functions using the characteristics (degree, end behavior, intercepts, and multiplicity of the zeros) of the function

Example 3: Graph the function $f(x)=3x^2-2x^3$



(1) Identify the **end behavior**:

So as $x\to-\infty$ $f(x)\to$ _____

$x\to\infty$ $f(x)\to$ _____

(2) Write the polynomial function in **factored form**.

(3) Find the **x-intercepts**: Set $f(x)=$ _____ and solve for _____:

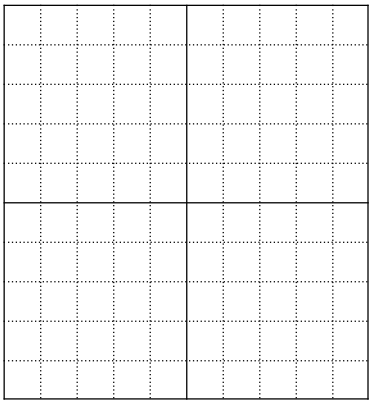
x-intercepts: $(___,___)$ *and* $(___,___)$

(4) Identify any **multiplicity** that occurs:

(5) **Find and plot points** between and beyond the x-intercepts:

(6) Draw the graph so that it passes through the plotted points and has appropriate end behavior and multiplicity. Check your answer by comparing your answer to your graphing calc.

Example 4: Graph the function $f(x)=\frac{2}{3}(x+1)^2(x-3)\left(x-\frac{9}{2}\right)$



(1) Identify the **end behavior**:

So as $x \to -\infty \quad f(x) \to$ _____

$x \to \infty \quad f(x) \to$ _____

(2) Write the polynomial function in **factored form**.

(4) Find the **x-intercepts**:

(7) Identify any **multiplicity** that occurs:

(8) **Find and plot points** between and beyond the x-intercepts:

(9) Draw the graph so that it passes through the plotted points and has appropriate end behavior and multiplicity. Check your answer by comparing your answer to your graphing calc.